Sheet 9

Unsharp joint readings

1. Let $\mathbf{n}_j = (\cos(2\pi j/3), \sin(2\pi j/3), 0)$ for $j = 0, 1, 2$. Show that $F_j = (\mathbb{1} + \mathbf{n}_j \cdot \boldsymbol{\sigma})/3$ is a POVM. Compute its square roots and Naimark isometry on $|0\rangle$; list the three probabilities. These are a concrete dilation computation, not another proof of the general theorem.
2. For $s, t = \pm 1$, test the proposed joint effects $M_{st} = [\mathbb{1} + \eta(s\sigma_x + t\sigma_z)]/4$, $0 \le \eta \le 1$. Compute spectra and marginals. For which η is this a joint POVM? Does the positivity test of this particular construction alone prove a necessity theorem about every possible joint reading?
3. **Phase-space matrix elements and moments (longer).** For $n, m \in \{0, 1, 2\}$ compute $\pi^{-1} \int_{\mathbb{C}} \langle h_n, \psi_\alpha\rangle\langle\psi_\alpha, h_m\rangle \, d^2\alpha$ from the coherent-state series, using polar coordinates. You may use $\int_0^\infty e^{-u} u^n \, du = n!$. This checks nine matrix elements, not the full resolution theorem, which is cited in L9. For coherent input $\beta = (1 + i)/\sqrt{2}$, find the mean physical position and momentum and their observed variances under the vacuum-generated phase-space POVM. Compare total and added spread products. State the parameter-port and ideal-detector assumptions of the eight-port implementation.

> **First hint**
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> Use polar coordinates in the coherent-state coefficients. Integrate the angular phase first to distinguish diagonal from off-diagonal matrix elements.

> **Further hint**
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> For a diagonal element, substitute the squared radius into the remaining radial integral and keep the polar Jacobian. For the physical moments, use the calibration factors in Lecture 9 and separate the input coherent-state spread from the extra spread of the reading before comparing their products.